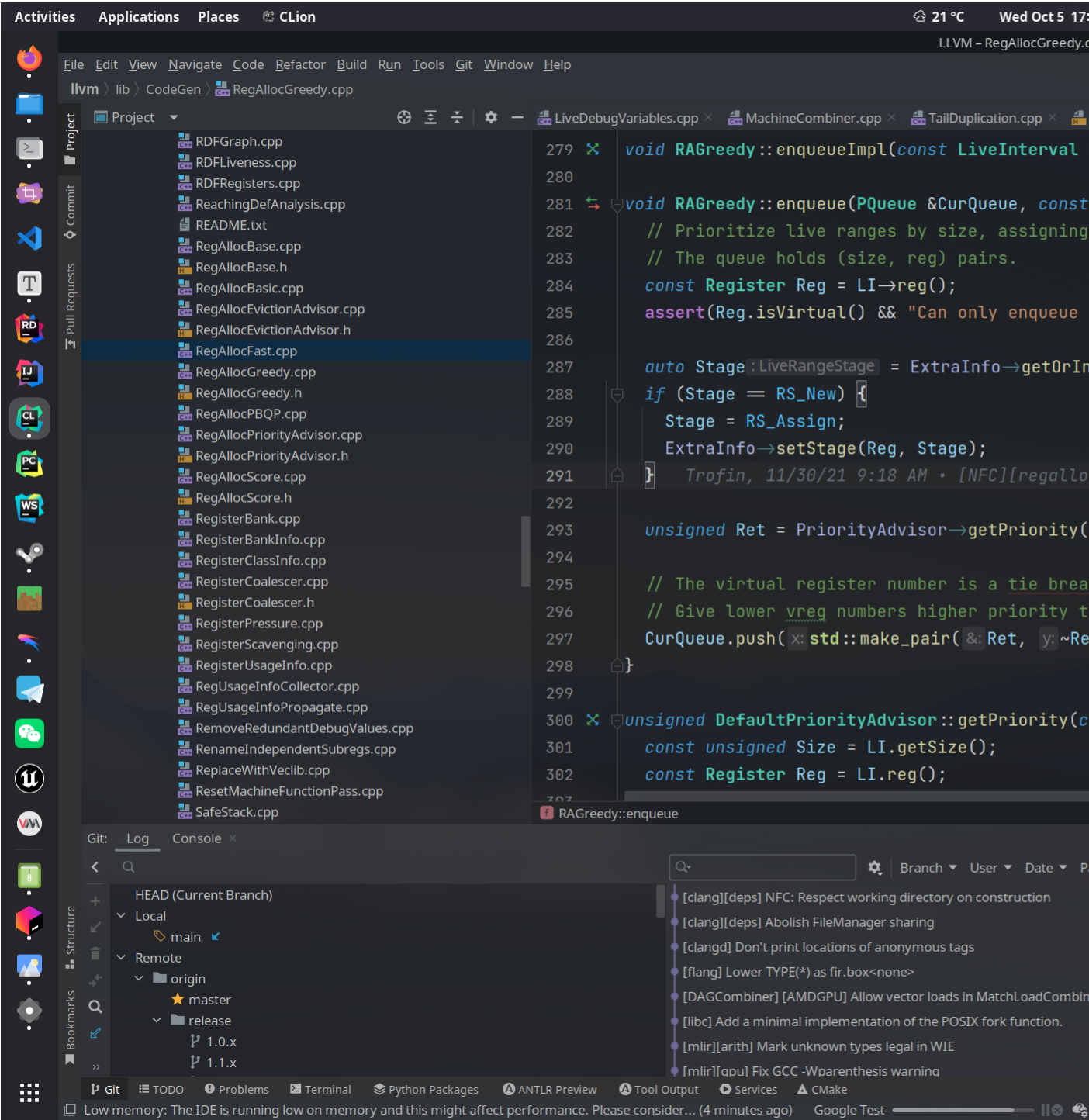


Clion

CLion

简介

CLion 是一款由 JetBrains 公司开发的功能丰富且强大的跨平台 C/C++ 集成开发环境 (IDE) 。



官方教程

在官方网站中给出了 [学习 CLion](#) 的教程。

安装

参见 [Download CLion](#)。

配置

工具链安装

CLion 默认不带编译器，构建工具和调试工具，需要手动进行安装。

Windows

参见 [Tutorial: Configure CLion on Windows | CLion Documentation](#)

值得一提的是 CLion 的 Windows 版本中自带了 MinGW，所以可以不用额外安装 MinGW 工具链。

Linux

Debian/Ubuntu 及其衍生发行版

```
1 sudo apt install make cmake # build tools
2 sudo apt install gcc g++ gdb # compiler and debugger
3 sudo apt install clang clang++ llvm lldb # you can also choose to use clang toolchain
```

Arch Linux 及其衍生发行版

```
1 sudo pacman -S make cmake # build tools
2 sudo pacman -S gcc g++ gdb # compiler and debugger
3 sudo pacman -S clang clang++ llvm lldb # you can also choose to use clang toolchain
```

Fedora/RHEL/CentOS/Rocky Linux

```
1 sudo dnf install make cmake # build tools
2 sudo dnf install gcc g++ gdb # compiler and debugger
3 sudo dnf install clang clang++ llvm lldb # you can also choose to use clang toolchain
```

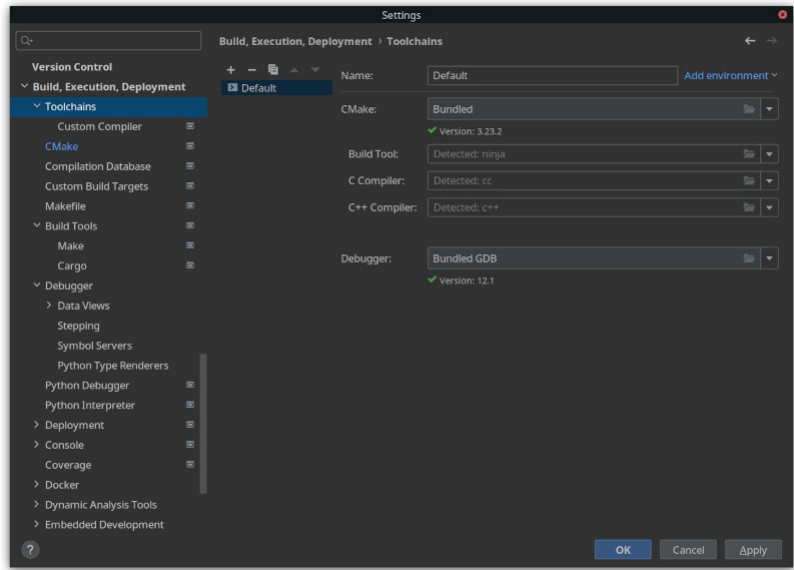
macOS

参见 [Tutorial: Configure CLion on macOS | CLion Documentation](#)

工具链设置

手动设置工具链

新安装的 CLion 会自动检测系统中的 C/C++ 开发工具链，如果已安装的工具链无法自动检测到，可在 Settings 中找到 Build, Execution, Deployment>Toolchains 进行手动配置。

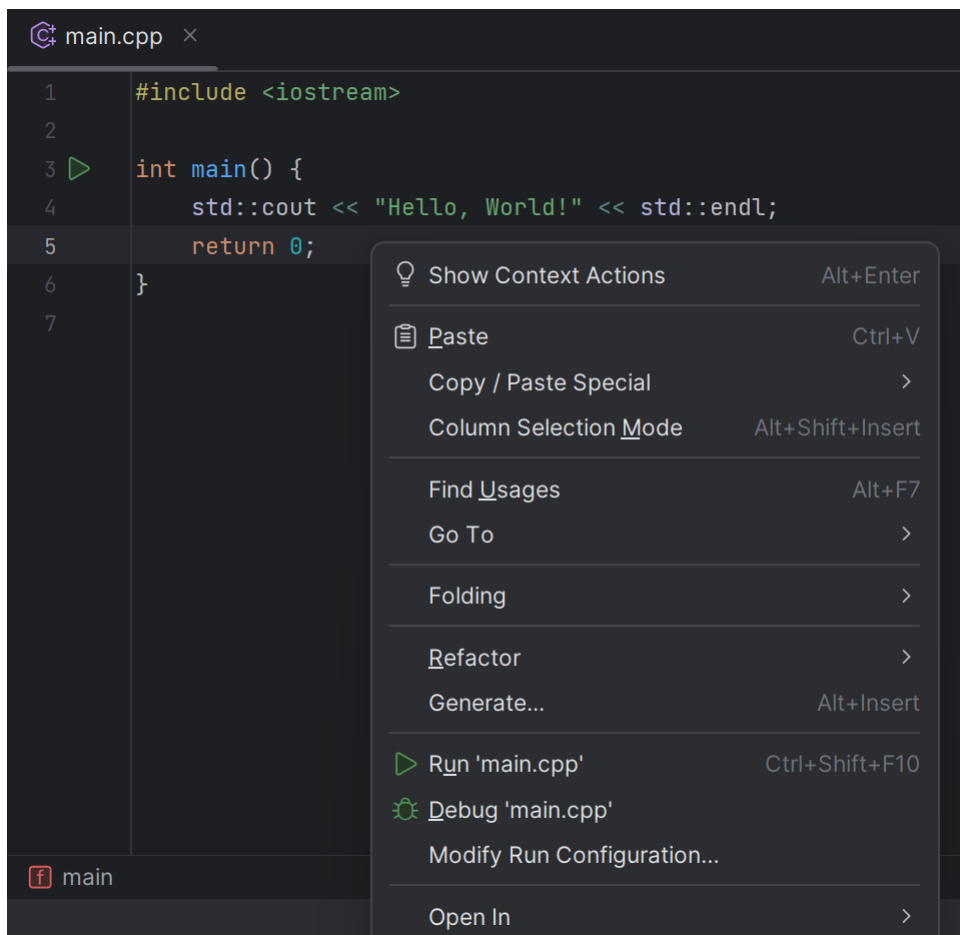


编译、运行和调试

虽然 CLion 诞生之初是面向多文件的复杂 C/C++ 项目诞生的，早些时候的 CLion 默认使用 CMake 作为构建工具，但是自 CLion 2022.3 版本起，CLion 已经支持 C, C++ 单文件运行。

有多种方式来运行一个 C++ 程序，一个简单的流程如下：

1. 创建一个 C/C++ 项目：New -> Project -> C++ Executable，选择合适的地址和语言标准版本，点击 Create。
2. 打开项目，此时的项目目录下应当存在一个 cmake-build-debug 目录、一个 CMakeLists.txt 文件和一个 main.cpp 文件。因为我们不需要使用 CMake 来管理项目，因此我们可以删去 CMakeLists.txt 文件和 cmake-build-debug 目录及其内所有文件。
3. 点击打开 main.cpp 文件，并在编辑区右键单击，可以看到 Run 'main.cpp' 选项。选择此选项后，CLion 可以自动创建一个运行配置并运行程序。

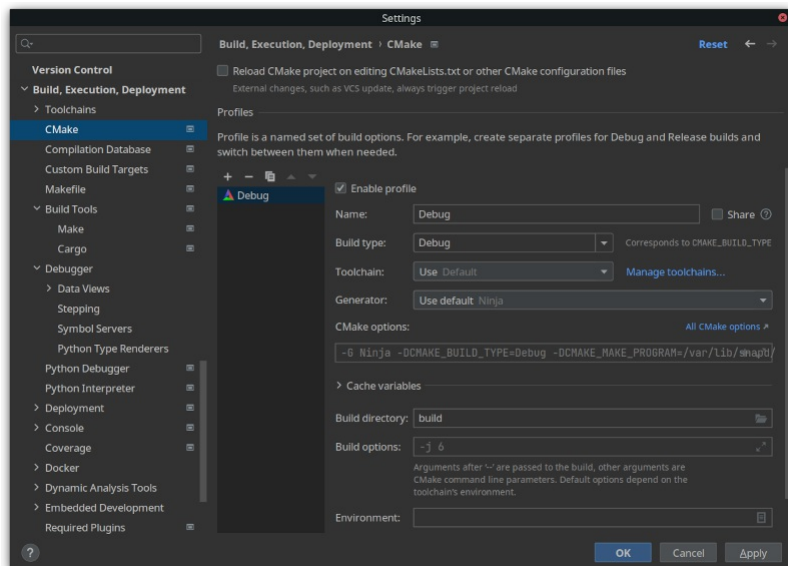


如需调试程序，可以编辑区打好断点，在编辑区右键单击，选择 `Debug 'main.cpp'` 选项。

通过 CMake 编译、运行和调试

设置

CLion 也可使用 [CMake](#) 作为构建工具，关于 CMake 的设置可以在 `Build, Execution, Deployment -> Toolchains -> CMake` 中修改。



编译选项

CMake 默认使用项目根目录下的 `CMakeList.txt` 作为构建项目的配置文件，可以使用 `add_compile_options` 命令来增加编译选项，例如：

```
1 add_compile_options(-std=c++17 -DDEBUG)
```

其他 CMake 的功能请参考 [CMake 官方文档](#)

免费获取 CLion IDE 许可证

CLion 为付费产品，但是可以通过教育邮箱或开源项目申请特殊许可证。申请之后不仅可以免费使用正版 CLion IDE，还可以免费使用 JetBrains 公司开发的其他付费产品。

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☒ Yes☐ No

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你可以到所在高校的教务中心官网去申请教育邮箱，如果申请不到需要使用 [学信网](#) 进行认证（仅中国大陆）。

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